

CS 2601 Linear and Convex Optimization

6. Gradient descent (part 2)

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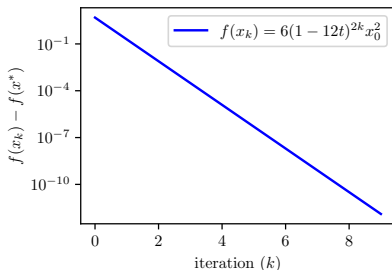
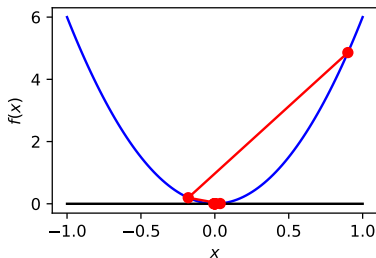
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Fast convergence

The following f is 12-smooth,

$$f(x) = 6x^2$$



For small enough step size t (e.g. 0.1),

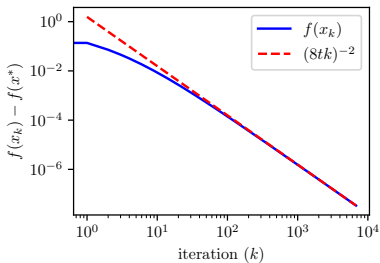
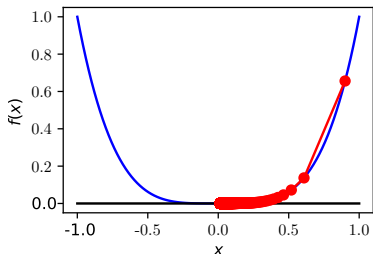
$$f(x_k) = 6x_0^2(1 - 12t)^{2k}$$

Need $O(\log \frac{1}{\epsilon})$ iterations to get within ϵ from optimal.

Slow convergence

The following f is also 12-smooth,

$$f(x) = \begin{cases} x^4, & \text{if } |x| \leq 1 \\ 4|x| - 3, & \text{if } |x| \geq 1 \end{cases}$$



For $x_0 \in (0, 1)$, small enough step size t (e.g. 0.1), and large k ,

$$x_k = \Theta\left(\frac{1}{\sqrt{k}}\right), \quad f(x_k) = \Theta\left(\frac{1}{k^2}\right)$$

Need $O(1/\sqrt{\epsilon})$ iterations to get within ϵ from optimal value (i.e. 0).

Strong convexity

A function f is **strongly convex** with parameter $m > 0$, or simply **m -strongly convex**, if

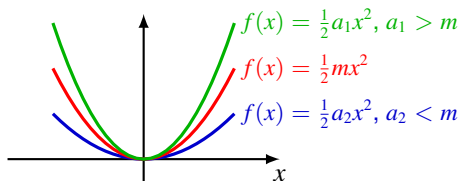
$$\tilde{f}(\mathbf{x}) = f(\mathbf{x}) - \frac{m}{2}\|\mathbf{x}\|^2$$

is convex.

Note. $f(\mathbf{x}) = \frac{m}{2}\|\mathbf{x}\|^2 + \tilde{f}(\mathbf{x})$, i.e. f is $\frac{m}{2}\|\mathbf{x}\|^2$ plus an extra convex term. Informally, “ m -strongly convex” means at least as “convex” as $\frac{m}{2}\|\mathbf{x}\|^2$.

Note. strong convexity $\implies$ strict convexity $\implies$ convexity

Example. $f(\mathbf{x}) = \frac{a}{2}\|\mathbf{x}\|^2$ is m -strongly convex iff $a \geq m$



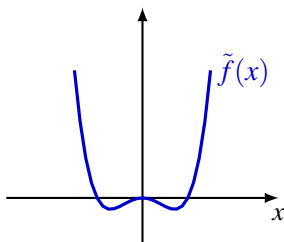
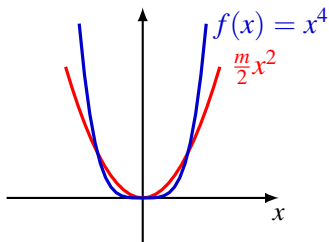
Strong convexity (cont'd)

Example. $f(\mathbf{x}) = \mathbf{a}^T \mathbf{x}$ is **not** m -strongly convex for any $m > 0$, as $\tilde{f}(\mathbf{x}) = \mathbf{a}^T \mathbf{x} - \frac{m}{2} \|\mathbf{x}\|^2$ is concave.

Example. $f(x) = x^4$ is **not** m -strongly convex for any $m > 0$, as $\tilde{f}(x) = x^4 - \frac{m}{2}x^2$ is not convex,

$$\tilde{f}''(x) = 12x^2 - m < 0$$

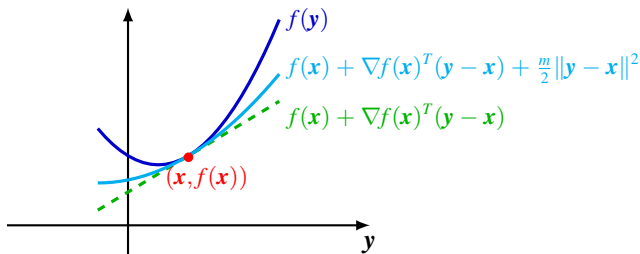
for $|x| < \sqrt{m/12}$.



First-order condition for strong convexity

A differentiable f is m -strongly convex iff

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T (\mathbf{y} - \mathbf{x}) + \frac{m}{2} \|\mathbf{x} - \mathbf{y}\|^2, \quad \forall \mathbf{x}, \mathbf{y}$$



- m -strong convexity and L -smoothness together imply

$$\frac{m}{2} \|\mathbf{x} - \mathbf{y}\|^2 \leq f(\mathbf{y}) - f(\mathbf{x}) - \nabla f(\mathbf{x})^T (\mathbf{y} - \mathbf{x}) \leq \frac{L}{2} \|\mathbf{x} - \mathbf{y}\|^2$$

Proof

1. By definition,

$$f \text{ is } m\text{-strongly convex} \iff \tilde{f}(\mathbf{x}) = f(\mathbf{x}) - \frac{m}{2}\|\mathbf{x}\|^2 \text{ is convex}$$

2. Note $\nabla \tilde{f}(\mathbf{x}) = \nabla f(\mathbf{x}) - m\mathbf{x}$. Check first-order condition for convexity,

$$\begin{aligned} & \tilde{f}(\mathbf{y}) - \tilde{f}(\mathbf{x}) - \nabla \tilde{f}(\mathbf{x})^T(\mathbf{y} - \mathbf{x}) \\ &= (f(\mathbf{y}) - \frac{m}{2}\|\mathbf{y}\|^2) - (f(\mathbf{x}) - \frac{m}{2}\|\mathbf{x}\|^2) - (\nabla f(\mathbf{x}) - m\mathbf{x})^T(\mathbf{y} - \mathbf{x}) \\ &= f(\mathbf{y}) - f(\mathbf{x}) - \nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x}) - \frac{m}{2}(\|\mathbf{y}\|^2 - \|\mathbf{x}\|^2 - 2\mathbf{x}^T(\mathbf{y} - \mathbf{x})) \\ &= f(\mathbf{y}) - f(\mathbf{x}) - \nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x}) - \frac{m}{2}\|\mathbf{x} - \mathbf{y}\|^2 \end{aligned}$$

Second-order condition for strong convexity

A twice continuously differentiable f is m -strongly convex iff

$$\nabla^2 f(\mathbf{x}) \succeq m\mathbf{I}, \quad \forall \mathbf{x}$$

or equivalently, the smallest eigenvalue of $\nabla^2 f(\mathbf{x})$ satisfies

$$\lambda_{\min}(\nabla^2 f(\mathbf{x})) \geq m, \quad \forall \mathbf{x}$$

Proof. $\tilde{f}(\mathbf{x}) = f(\mathbf{x}) - \frac{m}{2}\|\mathbf{x}\|^2$ is convex iff $\nabla^2 \tilde{f}(\mathbf{x}) = \nabla^2 f(\mathbf{x}) - m\mathbf{I} \succeq \mathbf{0}$

Example. With $\mathbf{Q} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$, we obtain $f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T \mathbf{Q} \mathbf{x} = \frac{1}{2}x_1^2 + x_2^2$ is 1-strongly convex.

More generally, $f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T \mathbf{Q} \mathbf{x}$ with $\mathbf{Q} \succ \mathbf{0}$ is $\lambda_{\min}(\mathbf{Q})$ -strongly convex, where $\lambda_{\min}(\mathbf{Q})$ is the smallest eigenvalue of $\mathbf{Q}$.

Bound on suboptimality gap

If f is m -strongly convex, then

$$f(\mathbf{x}) - f(\mathbf{x}^*) \leq \frac{1}{2m} \|\nabla f(\mathbf{x})\|^2$$

Note. When $\nabla f(\mathbf{x})$ is small, then $f(\mathbf{x})$ is close to optimal. (Do we have this property for general convex functions?)

Proof. By the first-order condition,

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T (\mathbf{y} - \mathbf{x}) + \frac{m}{2} \|\mathbf{y} - \mathbf{x}\|^2$$

Minimize over $\mathbf{y}$ on both sides,

$$f(\mathbf{x}^*) = \min_{\mathbf{y}} f(\mathbf{y}) \geq \min_{\mathbf{y}} \left[f(\mathbf{x}) + \nabla f(\mathbf{x})^T (\mathbf{y} - \mathbf{x}) + \frac{m}{2} \|\mathbf{y} - \mathbf{x}\|^2 \right]$$

The right-hand side is minimized at $\mathbf{y} = \mathbf{x} - \frac{1}{m} \nabla f(\mathbf{x})$, so

$$f(\mathbf{x}^*) \geq f(\mathbf{x}) - \frac{1}{2m} \|\nabla f(\mathbf{x})\|^2$$

Convergence: 1D example

$f(x) = \frac{1}{2}mx^2$ with $m > 0$ is both m -smooth and m -strongly convex..

Recall the gradient descent step is

$$x_{k+1} = x_k - tf'(x_k) = (1 - mt)x_k$$

and $x_k \rightarrow x^* = 0$ iff $t \in (0, \frac{2}{m})$.

If $t = \frac{1}{m}$, it gets to x^* in one step.

For $t \in (0, \frac{1}{m}) \cup (\frac{1}{m}, \frac{2}{m})$,

$$x_k = (1 - mt)^k x_0$$

so both $x_k \rightarrow x^*$ and $f(x_k) \rightarrow f(x^*)$ exponentially fast,

$$|x_k - x^*| = (1 - mt)^k \cdot |x_0 - x^*|$$

$$|f(x_k) - f(x^*)| = \frac{m(1 - mt)^{2k}}{2} |x_0 - x^*|^2$$

Convergence: Gradient flow

Consider the gradient flow,

$$\frac{d}{ds}\tilde{\mathbf{x}}_s = -\nabla f(\tilde{\mathbf{x}}_s)$$

If f is m -strongly convex and $\mathbf{x}^*$ is a minimum of f ,

$$\begin{aligned}\frac{d}{ds}\|\tilde{\mathbf{x}}_s - \mathbf{x}^*\|^2 &= 2\langle \tilde{\mathbf{x}}_s - \mathbf{x}^*, \frac{d}{ds}\tilde{\mathbf{x}}_s \rangle = 2\langle \nabla f(\tilde{\mathbf{x}}_s), \mathbf{x}^* - \tilde{\mathbf{x}}_s \rangle \\ &\leq 2[f(\mathbf{x}^*) - f(\tilde{\mathbf{x}}_s) - \frac{m}{2}\|\tilde{\mathbf{x}}_s - \mathbf{x}^*\|^2] \\ &\leq -m\|\tilde{\mathbf{x}}_s - \mathbf{x}^*\|^2\end{aligned}$$

Thus

$$\frac{d}{ds}[e^{ms}\|\tilde{\mathbf{x}}_s - \mathbf{x}^*\|^2] = e^{ms}\left[\frac{d}{ds}\|\tilde{\mathbf{x}}_s - \mathbf{x}^*\|^2 + m\|\tilde{\mathbf{x}}_s - \mathbf{x}^*\|^2\right] \leq 0$$

We obtain

$$\|\tilde{\mathbf{x}}_T - \mathbf{x}^*\|^2 \leq e^{-mT}\|\tilde{\mathbf{x}}_0 - \mathbf{x}^*\|^2$$

Convergence analysis

Theorem. If f is m -strongly convex and L -smooth, and $\mathbf{x}^*$ is a minimum of f , then for step size $t \in (0, \frac{1}{L}]$, the sequence $\{\mathbf{x}_k\}$ produced by the gradient descent algorithm satisfies

$$f(\mathbf{x}_k) - f(\mathbf{x}^*) \leq (1 - mt)^k [f(\mathbf{x}_0) - f(\mathbf{x}^*)]$$

$$\|\mathbf{x}_k - \mathbf{x}^*\|^2 \leq (1 - mt)^k \|\mathbf{x}_0 - \mathbf{x}^*\|^2$$

Notes.

- $0 \leq 1 - \frac{m}{L} \leq 1 - mt < 1$, so $\mathbf{x}_k \rightarrow \mathbf{x}^*$ and $f(\mathbf{x}_k) \rightarrow f(\mathbf{x}^*)$ exponentially fast
- The number of iterations to reach $f(\mathbf{x}_k) - f(\mathbf{x}^*) \leq \epsilon$ is $O(\log \frac{1}{\epsilon})$. For $\epsilon = 10^{-p}$, $k = O(p)$, linear in the number of significant digits!
- Since $\nabla f(\mathbf{x}^*) = 0$, the bounds on slide 5 yield

$$\frac{m}{2} \|\mathbf{x}_k - \mathbf{x}^*\|^2 \leq f(\mathbf{x}_k) - f(\mathbf{x}^*) \leq \frac{L}{2} \|\mathbf{x}_k - \mathbf{x}^*\|^2$$

relating bounds on $\|\mathbf{x}_k - \mathbf{x}^*\|^2$ and those on $f(\mathbf{x}_k) - f(\mathbf{x}^*)$

Proof

For the bound on $\|\mathbf{x}_k - \mathbf{x}^*\|$, similar to the proof without strong convexity, with difference highlighted.

1. By the basic gradient step $\mathbf{x}_{k+1} = \mathbf{x}_k - t\nabla f(\mathbf{x}_k)$,

$$\begin{aligned}\|\mathbf{x}_{k+1} - \mathbf{x}^*\|^2 - \|\mathbf{x}_k - \mathbf{x}^*\|^2 &= \|\mathbf{x}_k - t\nabla f(\mathbf{x}_k) - \mathbf{x}^*\|^2 - \|\mathbf{x}_k - \mathbf{x}^*\|^2 \\ &= 2t\nabla f(\mathbf{x}_k)^T(\mathbf{x}^* - \mathbf{x}_k) + t^2\|\nabla f(\mathbf{x}_k)\|^2\end{aligned}$$

2. By m -strong convexity,

$$\nabla f(\mathbf{x}_k)^T(\mathbf{x}^* - \mathbf{x}_k) \leq f(\mathbf{x}^*) - f(\mathbf{x}_k) - \frac{m}{2}\|\mathbf{x}_k - \mathbf{x}^*\|^2$$

3. By L -smoothness, the second term is upper bounded by

$$t^2\|\nabla f(\mathbf{x}_k)\|^2 \leq 2t[f(\mathbf{x}_k) - f(\mathbf{x}_{k+1})]$$

4. Plugging 2 and 3 into 1,

$$\|\mathbf{x}_{k+1} - \mathbf{x}^*\|^2 \leq (1 - mt)\|\mathbf{x}_k - \mathbf{x}^*\|^2 + 2t[f(\mathbf{x}^*) - f(\mathbf{x}_{k+1})]$$

5. Since $f(\mathbf{x}^*) \leq f(\mathbf{x}_{k+1})$,

$$\|\mathbf{x}_{k+1} - \mathbf{x}^*\|^2 \leq (1 - mt)\|\mathbf{x}_k - \mathbf{x}^*\|^2$$

Proof

For the bound on $f(\mathbf{x}_k) - f(\mathbf{x}^*)$,

1. By L -smoothness, lower bound improvement in the k -th iteration,

$$f(\mathbf{x}_k) - f(\mathbf{x}_{k+1}) \geq \frac{t}{2} \|\nabla f(\mathbf{x}_k)\|^2$$

2. By m -strong convexity, upper bound suboptimality gap (slide 8),

$$f(\mathbf{x}_k) - f(\mathbf{x}^*) \leq \frac{1}{2m} \|\nabla f(\mathbf{x}_k)\|^2$$

3. Eliminate $\|\nabla f(\mathbf{x}_k)\|$ from the above two inequalities,

$$f(\mathbf{x}_{k+1}) - f(\mathbf{x}^*) \leq (1 - mt) [f(\mathbf{x}_k) - f(\mathbf{x}^*)]$$

so

$$f(\mathbf{x}_{k+1}) - f(\mathbf{x}^*) \leq (1 - mt)^k [f(\mathbf{x}_0) - f(\mathbf{x}^*)]$$

Convergence: 2D quadratic function

$$f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T \mathbf{Q} \mathbf{x}, \quad \mathbf{Q} = \begin{bmatrix} m & 0 \\ 0 & L \end{bmatrix}$$

where $L > m > 0$. f is L -smooth and m -strongly convex. $\mathbf{x}^* = \mathbf{0}$.

The gradient descent step is

$$\mathbf{x}_{k+1} = \mathbf{x}_k - t \nabla f(\mathbf{x}_k) = (\mathbf{I} - t\mathbf{Q})\mathbf{x}_k$$

so

$$\mathbf{x}_k = (\mathbf{I} - t\mathbf{Q})^k \mathbf{x}_0 = \begin{bmatrix} (1 - mt)^k x_{01} \\ (1 - Lt)^k x_{02} \end{bmatrix}$$

and

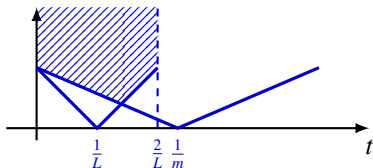
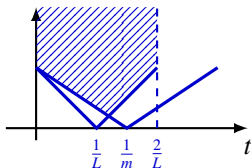
$$f(\mathbf{x}_k) = \frac{m}{2}(1 - mt)^{2k} x_{01}^2 + \frac{L}{2}(1 - Lt)^{2k} x_{02}^2$$

To ensure convergence, $t < \frac{2}{L}$. The convergence rate is determined by the slower of $(1 - Lt)^{2k}$ and $(1 - mt)^{2k}$.

Convergence: 2D quadratic function (cont'd)

To maximize convergence rate, solve

$$\begin{aligned} \min_t \quad & \max\{|1 - Lt|, |1 - mt|\} \\ \text{s. t.} \quad & 0 < t < 2/L \end{aligned}$$



Maximum rate achieved by $1 - mt = Lt - 1 \implies t = \frac{2}{m+L}$, in which case

$$\mathbf{x}_k = \left(\frac{L-m}{L+m}\right)^k \begin{bmatrix} x_{01} \\ (-1)^k x_{02} \end{bmatrix} \implies \|\mathbf{x}_k - \mathbf{x}^*\|_2 = \left(\frac{L-m}{L+m}\right)^k \|\mathbf{x}_0 - \mathbf{x}^*\|_2$$

$$f(\mathbf{x}_k) - f(\mathbf{x}^*) = \left(\frac{L-m}{L+m}\right)^{2k} [f(\mathbf{x}_0) - f(\mathbf{x}^*)]$$

Depends on $\kappa(\mathbf{Q}) = \frac{\lambda_{\max}(\mathbf{Q})}{\lambda_{\min}(\mathbf{Q})} = \frac{L}{m}$, the **condition number** of $\mathbf{Q}$

Condition number

For a matrix $\mathbf{Q} \in \mathbb{R}^{n \times n}$ s.t. $\mathbf{Q} \succ \mathbf{0}$, its **condition number**¹ is defined as

$$\kappa(\mathbf{Q}) = \frac{\lambda_{\max}(\mathbf{Q})}{\lambda_{\min}(\mathbf{Q})}$$

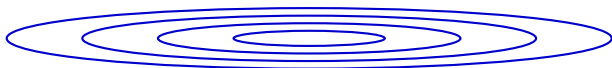
It characterizes how stretched the level curves of $f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T \mathbf{Q} \mathbf{x}$ are.

Example. $\mathbf{Q} = \text{diag}\{\gamma, 1\}$, $f(x_1, x_2) = \frac{\gamma}{2}x_1^2 + \frac{1}{2}x_2^2$



$$\mathbf{Q} = \text{diag}\{1, 1\}$$

$$\kappa(\mathbf{Q}) = 1$$



$$\mathbf{Q} = \text{diag}\{0.01, 1\}$$

$$\kappa(\mathbf{Q}) = 100$$

Nondiagonal case reduces to diagonal case in eigenbasis of $\mathbf{Q}$.

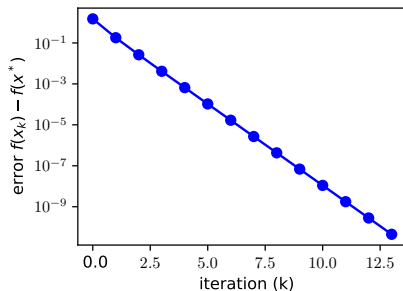
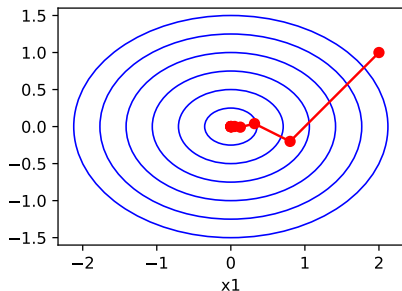
For nonquadratic case, $\kappa(\nabla^2 f(\mathbf{x}))$ plays a similar role.

¹For a general nonsingular matrix, the condition number is the ratio between its largest and smallest singular values, $\kappa(\mathbf{A}) = \sigma_{\max}(\mathbf{A})/\sigma_{\min}(\mathbf{A})$.

Well-conditioned Problem

The problem $\min_x \frac{1}{2} \mathbf{x}^T \mathbf{Q} \mathbf{x}$ is **well-conditioned** if $\kappa(\mathbf{Q})$ is small.

Example. $\mathbf{Q} = \text{diag}\{0.5, 1\}$, $f(x_1, x_2) = \frac{1}{4}x_1^2 + \frac{1}{2}x_2^2$, $\kappa(\mathbf{Q}) = 2$



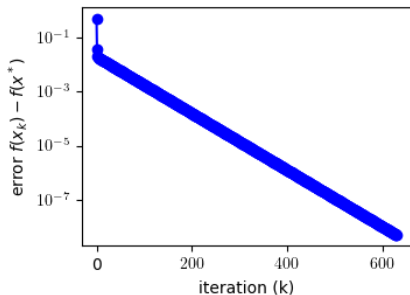
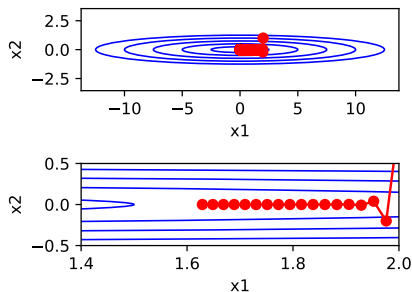
Fast convergence: for $\mathbf{x}_0 = (2, 1)$, $t = 1.2$, and large k ,

$$f(\mathbf{x}_k) \sim \frac{m}{2}(1 - mt)^{2k}x_{01}^2 = (0.4)^{2k}$$

Ill-conditioned problem

The problem $\min_x \frac{1}{2} \mathbf{x}^T \mathbf{Q} \mathbf{x}$ is **ill-conditioned** if $\kappa(\mathbf{Q})$ is large.

Example. $\mathbf{Q} = \text{diag}\{0.01, 1\}$, $f(x_1, x_2) = \frac{1}{200}x_1^2 + \frac{1}{2}x_2^2$, $\kappa(\mathbf{Q}) = 100$



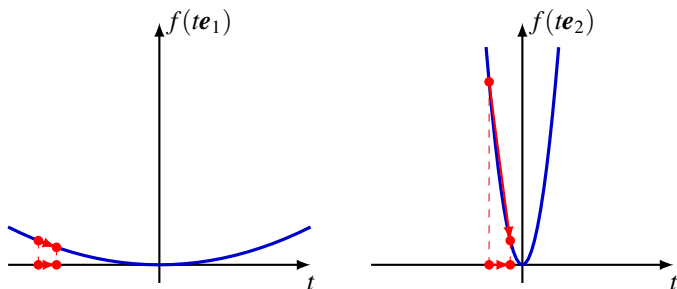
Slow convergence (relatively): for $\mathbf{x}_0 = (2, 1)$, $t = 1.2$, and large k ,

$$f(\mathbf{x}_k) \sim \frac{m}{2} (1 - mt)^{2k} x_{01}^2 = \frac{1}{50} (0.988)^{2k}$$

III-conditioned problem (cont'd)

$$f(x_1, x_2) = \frac{1}{2} \mathbf{x}^T \mathbf{Q} \mathbf{x} = \frac{1}{200} x_1^2 + \frac{1}{2} x_2^2, \quad \mathbf{Q} = \text{diag}\{0.01, 1\}, \quad \kappa(\mathbf{Q}) = 100$$

- 1-smooth $\implies$ To guarantee convergence, step size² $t < 2$
- This limit is imposed by movement along e_2 direction
- Too pessimistic along other directions, e.g. along e_1 , can use $t < 200$



²We proved convergence for $t \in (0, 1/L]$. The proofs can be modified slightly to show convergence for $t \in (0, 2/L]$.

Ill-condition problem (cont'd)

The negative gradient direction is far away from the “ideal” direction for ill-conditioned problem.

For $\mathbf{Q} = \text{diag}\{\gamma, 1\}$, $f(x_1, x_2) = \frac{1}{2}\mathbf{x}^T \mathbf{Q} \mathbf{x} = \frac{\gamma}{2}x_1^2 + \frac{1}{2}x_2^2$,

negative gradient direction

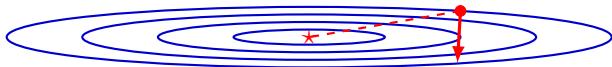
$$-\nabla f(\mathbf{x}) = -\mathbf{Q}\mathbf{x} = (-\gamma x_1, -x_2)$$

“ideal” direction

$$\mathbf{d} = -\mathbf{x}$$



$$\mathbf{Q} = \text{diag}\{1, 1\}$$
$$\kappa(\mathbf{Q}) = 1$$



$$\mathbf{Q} = \text{diag}\{0.01, 1\}$$
$$\kappa(\mathbf{Q}) = 100$$